

2520090088(B.DHARSHAN)

SKILL 13

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root@PC: ~/os_lab/file_redir_lab
root@PC:~# cd ~/os_lab
mkdir -p file_redir_lab && cd file_redir_lab
root@PC:~/os_lab/file_redir_lab# nano input_redir.c
root@PC:~/os_lab/file_redir_lab# nano output_redir.c
root@PC:~/os_lab/file_redir_lab# gcc -g -O0 -Wall -pthread output_redir.c -o output_redir
./output_redir
=== Task 2: Output Redirection Test (>, >>, Permissions) ===

--- Test 2.1: Overwrite Redirection (echo 'Line 1' > output_test.txt) ---
--- Contents of output_test.txt ---
Line 1: Overwrite Mode Active
-----

--- Test 2.2: Append Redirection (echo 'Line 2' >> output_test.txt) ---
--- Contents of output_test.txt ---
Line 1: Overwrite Mode Active
Line 2: Appended Entry
-----

--- Test 2.3: Permission Denied Test ---
root@PC:~/os_lab/file_redir_lab# valgrind --leak-check=full --track-fds=yes ./input_redir
valgrind: ./input_redir: No such file or directory
==566== Memcheck, a memory error detector
==566== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==566== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==566== Command: ./output_redir
==566==
=== Task 2: Output Redirection Test (>, >>, Permissions) ===

--- Test 2.1: Overwrite Redirection (echo 'Line 1' > output_test.txt) ---
--- Contents of output_test.txt ---
Line 1: Overwrite Mode Active
-----

--- Test 2.2: Append Redirection (echo 'Line 2' >> output_test.txt) ---
--- Contents of output_test.txt ---
Line 1: Overwrite Mode Active
```

root@PC: ~/os_lab/file_redir_lab

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--- Test 2.2: Append Redirection (echo 'Line 2' >> output_test.txt) ---
--- Contents of output_test.txt ---
Line 1: Overwrite Mode Active
Line 2: Appended Entry
-----

--- Test 2.3: Permission Denied Test ---
==566==
==566== FILE DESCRIPTORS: 3 open (3 std) at exit.
==566==
==566== HEAP SUMMARY:
==566==   in use at exit: 0 bytes in 0 blocks
==566==   total heap usage: 5 allocs, 5 frees, 10,160 bytes allocated
==566==
==566== All heap blocks were freed -- no leaks are possible
==566==
==566== For lists of detected and suppressed errors, rerun with: -s
==566== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
root@PC:~/os_lab/file_redir_lab# strace -f -e trace=openat,dup2,close ./output_redir
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
close(3)                                = 0
=== Task 2: Output Redirection Test (>, >>, Permissions) ===

--- Test 2.1: Overwrite Redirection (echo 'Line 1' > output_test.txt) ---
openat(AT_FDCWD, "output_test.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
dup2(4, 1)                                = 1
close(4)                                = 0
strace: Process 574 attached
[pid 574] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4)                                = 0
[pid 574] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC)
= 4
[pid 574] close(4)                                = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) =
-1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) =
4
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root@PC: ~/os_lab/file_redir_lab

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OEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NAME", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC)
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root@PC: ~/os_lab/file_redir_lab
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 574] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 4
[pid 574] close(4) = 0
[pid 574] close(1) = 0
[pid 574] close(2) = 0
[pid 574] +++ exited with 0 +++
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=574, si_uid=0, si_status=0, si_utime=0, si_stime=1 /* 0.01 s */} ---
dup2(3, 1) = 1
close(3) = 0
--- Contents of output_test.txt ---
openat(AT_FDCWD, "output_test.txt", O_RDONLY) = 3
Line 1: Overwrite Mode Active
close(3) = 0
-----

--- Test 2.2: Append Redirection (echo 'Line 2' >> output_test.txt) ---
openat(AT_FDCWD, "output_test.txt", O_WRONLY|O_CREAT|O_APPEND, 0644) = 4
dup2(4, 1) = 1
close(4) = 0
strace: Process 575 attached
[pid 575] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 4

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root@PC: ~/os_lab/file_redir_lab

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--- Test 2.2: Append Redirection (echo 'Line 2' >> output_test.txt) ---
openat(AT_FDCWD, "output_test.txt", O_WRONLY|O_CREAT|O_APPEND, 0644) = 4
dup2(4, 1)                                = 1
close(4)                                  = 0
strace: Process 575 attached
[pid 575] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4)                                = 0
[pid 575] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4)                                = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4)                                = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4)                                = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4)                                = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4)                                = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4)                                = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4)                                = 0
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root@PC: ~/os_lab/file_redir_lab
[pid 575] close(4) = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC)
= 4
[pid 575] close(4) = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEX
EC) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXE
C) = 4
[pid 575] close(4) = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O
_RDONLY|O_CLOEXEC) = 4
[pid 575] close(4) = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEX
EC) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXE
C) = 4
[pid 575] close(4) = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXE
C) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC
) = 4
[pid 575] close(4) = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) =
4
[pid 575] close(4) = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXE
C) = -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC
) = 4
[pid 575] close(4) = 0
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
[pid 575] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC)
= 4
[pid 575] close(4) = 0
[pid 575] close(1) = 0
```

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root@PC: ~/os_lab/file_redir_lab
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] close(1) = 0
[pid 576] close(2) = 0
[pid 576] +++ exited with 0 +++
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=576, si_uid=0, si_status=0, si_etime=0, si_stime=2 /* 0.02 s */} ---
dup2(3, 1) = 1
close(3) = 0
+++ exited with 0 +++
root@PC:~/os_lab/file_redir_lab#
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root@PC: ~/os_lab/file_redir_lab
openat(AT_FDCWD, "readonly_test.txt", O_WRONLY|O_CREAT|O_TRUNC, 0444) = 3
close(3) = 0
openat(AT_FDCWD, "readonly_test.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
dup2(4, 1) = 1
close(4) = 0
strace: Process 576 attached
[pid 576] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 4
[pid 576] close(4) = 0
[pid 576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC)
```